Missingness Report

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Contents

1 Read Data

2 Order of Events

2.1 event, study pairs

```
all |> dplyr::distinct(event, study)
```

```
## # A tibble: 11 x 2
##    event study
##    <chr> <chr>
##  1 V1    HCA
##  2 F1    HCA
##  3 V2    HCA
##  4 F2    HCA
##  5 CR    HCA
##  6 V3    AABC
##  7 F3    HCA
##  8 V2    AABC
##  9 AF1   AABC
## 10 V4    AABC
## 11 V1    AABC
```

- notice some peoples first visit (`V1`) may be in `HCA` or, `AABC` – same being true for `V2`
  - suggesting people joined study at different time periods

```
all <- all |>
  dplyr::mutate(
    event = factor(
      event,
      levels = c(
        "V1", "F1",
        "V2", "F2",
        "CR",
        "V3", "F3",
        "V4",
        "AF1"
      ),
      ordered = TRUE
```

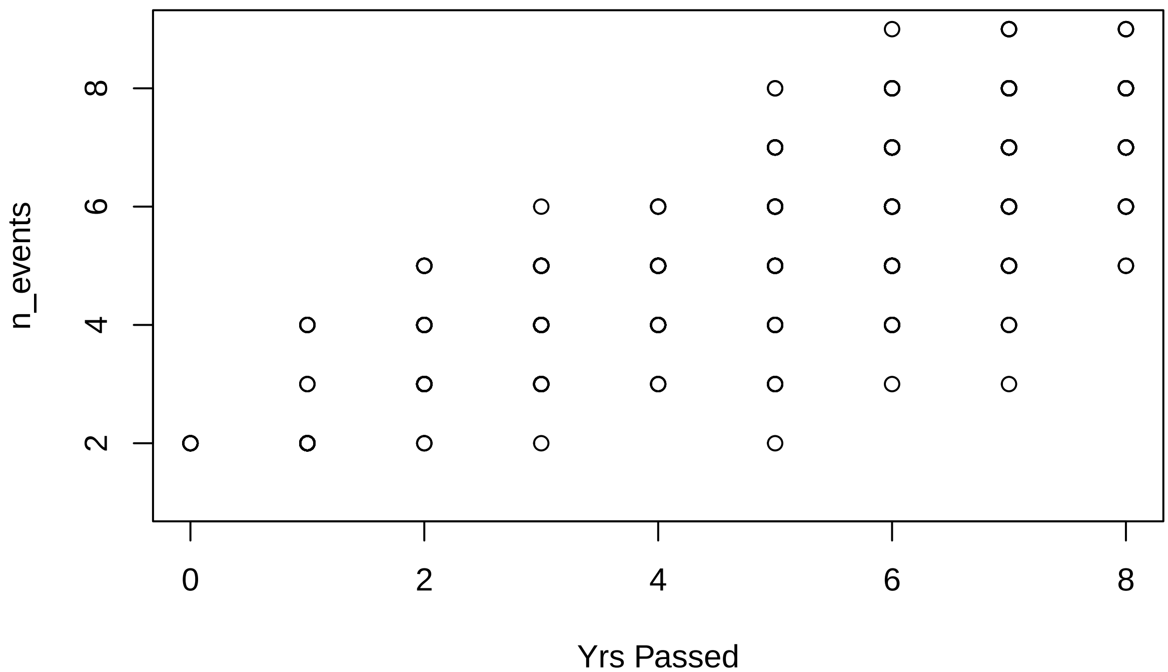
```r
    ),

    study = factor(
      study,
      levels = c("HCA", "AABC"),
      ordered = TRUE
    )
  )
```

```r
patient_paths <-
  all |>
  filter(age_open != "90 or older") |>
  mutate(age_open = as.numeric(age_open)) |>
  mutate(
    patient = str_remove(id, "^[A-Z]+")
  ) |>
  distinct(patient, study, event, age_open) |>
  arrange(patient, study, event) |>
  group_by(patient) |>
  arrange(study, event) |>
  summarise(
    n_studies = n_distinct(study),
    n_events = n(),

    years_passed = if_else(
      n() > 1,
      max(age_open, na.rm = TRUE) - min(age_open, na.rm = TRUE),
      NA_real_
    ),

    study_event_path = paste(
      paste(study, event, sep = ":"),
      collapse = " → "
    ),

    .groups = "drop"
  ) |>
  select(patient, years_passed, study_event_path, years_passed, everything()) |>
  arrange(desc(years_passed))
patient_paths
```

```
## # A tibble: 1,350 x 5
##    patient years_passed study_event_path                      n_studies n_events
##    <chr>          <dbl> <chr>                                     <int>    <int>
##  1 6010538            8 HCA:V1 → HCA:F1 → HCA:F2 → HCA:CR → ~         2        8
##  2 6042046            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F3 → ~         2        5
##  3 6131449            8 HCA:V1 → HCA:F1 → HCA:F2 → HCA:CR → ~         2        7
##  4 6228767            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        8
##  5 6249977            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        9
##  6 6367781            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        8
##  7 6397992            8 HCA:V1 → HCA:F1 → HCA:F2 → HCA:CR → ~         2        8
##  8 6399087            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        7
##  9 6429272            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        8
```

```
## 10 6461066            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        9
## # i 1,340 more rows
```

```r
plot(patient_paths$years_passed, patient_paths$n_events, xlab = "Yrs Passed", ylab = "n_events")
```



3 Path Diagram

```r
patient_paths
```

```
## # A tibble: 1,350 x 5
##    patient years_passed study_event_path                      n_studies n_events
##    <chr>          <dbl> <chr>                                     <int>    <int>
##  1 6010538            8 HCA:V1 → HCA:F1 → HCA:F2 → HCA:CR → ~         2        8
##  2 6042046            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F3 → ~         2        5
##  3 6131449            8 HCA:V1 → HCA:F1 → HCA:F2 → HCA:CR → ~         2        7
##  4 6228767            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        8
##  5 6249977            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        9
##  6 6367781            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        8
##  7 6397992            8 HCA:V1 → HCA:F1 → HCA:F2 → HCA:CR → ~         2        8
##  8 6399087            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        7
##  9 6429272            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        8
## 10 6461066            8 HCA:V1 → HCA:F1 → HCA:V2 → HCA:F2 → ~         2        9
## # i 1,340 more rows
```

```r
library(tidyverse)

patient_steps <-
  patient_paths |>
  separate_rows(study_event_path, sep = " → ") |>
  separate(
    study_event_path,
    into = c("study", "event"),
    sep = ":"
  ) |>
  mutate(
    study = factor(
      study,
      levels = c("HCA", "AABC"),
      ordered = TRUE
    ),
    event = factor(
      event,
      levels = c(
        "V1", "F1",
        "V2", "F2",
        "CR",
        "V3", "F3",
        "V4",
        "AF1"
      ),
      ordered = TRUE
    )
  ) |>
  arrange(patient, study, event) |>
  group_by(patient) |>
  mutate(step = row_number()) |>
  ungroup()
```

```r
patient_steps <-
  patient_steps |>
  mutate(
    study_event = interaction(
      study,
      event,
      sep = ":",
      lex.order = TRUE
    )
  )
```

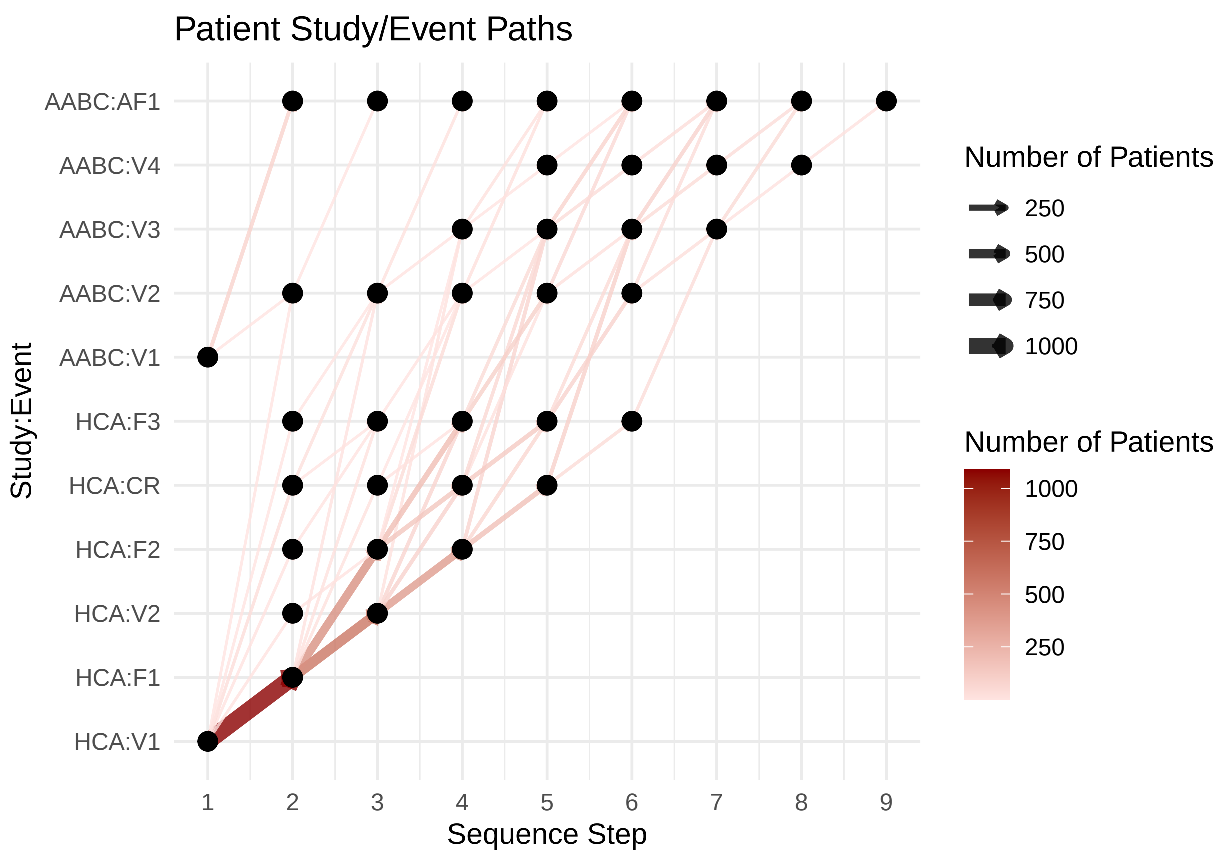
```r
patient_edges <-
  patient_steps |>
  arrange(patient, step) |>
  group_by(patient) |>
  mutate(
    next_study_event = lead(study_event),
    next_step = lead(step)
  ) |>
```

```r
  filter(!is.na(next_study_event)) |>
  ungroup()
```

```r
transition_counts <-
  patient_edges |>
  count(
    step,
    next_step,
    study_event,
    next_study_event,
    name = "n_patients"
  )
```

```r
library(grid)

max_step <- max(patient_steps$step)

path_plot <-
  ggplot(transition_counts) +
  geom_segment(
    aes(
      x = step,
      xend = next_step,
      y = study_event,
      yend = next_study_event,
      color = n_patients,
      linewidth = n_patients
    ),
    arrow = arrow(length = unit(0.15, "cm")),
    alpha = 0.8
  ) +
  geom_point(
    data = patient_steps |> distinct(step, study_event),
    aes(x = step, y = study_event),
    size = 3
  ) +
  scale_x_continuous(
    breaks = seq(1, max_step, by = 1),
    limits = c(1, max_step)
  ) +
  scale_linewidth(range = c(0.5, 3)) +
  scale_color_gradient(
    low = "mistyrose",
    high = "red4"
  ) +
  labs(
    x = "Sequence Step",
    y = "Study:Event",
    color = "Number of Patients",
    linewidth = "Number of Patients",
    title = "Patient Study/Event Paths"
  ) +
  theme_minimal()
```

```
path_plot
```



4 Missingness Relevance

A great deal of data is missing (`NA`) – therefore our data is heavily **biased**. We would like to determine in *what way* its **biased.** Perhaps by adjusting for this bias we see strong associations for each variable to the cognitive markers in studying dimentia.

4.1 Blood Data

```
library(dplyr)
library(naniar)
blood |>
  miss_var_summary() |>
  as.data.frame() |>
  select(variable, pct_miss)
```

```
##             variable pct_miss
## 1         direct_ldl     100
## 2        aldosterone      96.5
## 3               IL13      93.1
```

```
## 4                IL2      93.1
## 5              IL17A      93.0
## 6               CCL8      92.7
## 7            TNFSF10      92.7
## 8                IL4      92.5
## 9               IL33      92.3
## 10             CCL13      92.2
## 11             CCL19      92.2
## 12             IL17F      92.1
## 13               EGF      92.1
## 14            CXCL12      92.1
## 15              OLR1      92.1
## 16              IL27      92.1
## 17             CXCL9      92.1
## 18              TGFA      92.1
## 19              IL1B      92.1
## 20               IL6      92.1
## 21           TNFSF12      92.1
## 22             CCL11      92.1
## 23               HGF      92.1
## 24            FLT3LG      92.1
## 25               IL7      92.1
## 26              IL18      92.1
## 27            CXCL10      92.1
## 28              IFNG      92.1
## 29              IL10      92.1
## 30               TNF      92.1
## 31              IL15      92.1
## 32              CCL3      92.1
## 33             CXCL8      92.1
## 34             MMP12      92.1
## 35              CSF2      92.1
## 36              CSF3      92.1
## 37             VEGFA      92.1
## 38             IL17C      92.1
## 39              CCL2      92.1
## 40               OSM      92.1
## 41              CSF1      92.1
## 42              CCL4      92.1
## 43               LTA      92.1
## 44              CCL7      92.1
## 45              MMP1      92.1
## 46 pT217_Conc_pg_ml      88.2
## 47  GFAP_Conc_pg_ml      88.1
## 48   NfL_Conc_pg_ml      88.1
## 49  tTau_Conc_pg_ml      88.1
## 50             dheas      85.4
## 51          cortisol      85.2
## 52         estradiol      62.3
## 53      testosterone      61.6
## 54    totalbilirubin      60.8
## 55             hscrp      60.4
## 56                lh      60.4
## 57               fsh      60.4
```

```
## 58            hba1c     60.3
## 59   friedewald_ldl     60.3
## 60         alt_sgpt     60.2
## 61         ast_sgot     60.2
## 62         vitamind     60.2
## 63    alkphos_total     60.2
## 64          calcium     60.2
## 65        potassium     60.2
## 66          albumin     60.2
## 67         chloride     60.2
## 68       co2content     60.2
## 69       creatinine     60.2
## 70          glucose     60.2
## 71           sodium     60.2
## 72     totalprotein     60.2
## 73     ureanitrogen     60.2
## 74              hdl     60.2
## 75      cholesterol     60.2
## 76     triglyceride     60.2
## 77          insulin     60.2
## 78         id_event      0
## 79               id      0
## 80            event      0
## 81            study      0
```

- direct_ldl is entirely missing, we may discard this feature
- Many repeated values

4.2 Diet Data

```
diet |>
  miss_var_summary() |>
  as.data.frame() |>
  select(variable, pct_miss)
```

```
##                   variable pct_miss
## 1           asa24_numfoods     84.6
## 2           asa24_numcodes     84.6
## 3               asa24_kcal     84.6
## 4               asa24_prot     84.6
## 5               asa24_tfat     84.6
## 6               asa24_carb     84.6
## 7               asa24_mois     84.6
## 8                asa24_alc     84.6
## 9               asa24_caff     84.6
## 10              asa24_theo     84.6
## 11              asa24_sugr     84.6
## 12              asa24_fibe     84.6
## 13              asa24_calc     84.6
## 14              asa24_iron     84.6
## 15              asa24_magn     84.6
```

```
## 16          asa24_phos    84.6
## 17          asa24_pota    84.6
## 18          asa24_sodi    84.6
## 19          asa24_zinc    84.6
## 20          asa24_copp    84.6
## 21          asa24_sele    84.6
## 22            asa24_vc    84.6
## 23           asa24_vb1    84.6
## 24           asa24_vb2    84.6
## 25          asa24_niac    84.6
## 26           asa24_vb6    84.6
## 27          asa24_fola    84.6
## 28            asa24_fa    84.6
## 29            asa24_ff    84.6
## 30          asa24_fdfe    84.6
## 31          asa24_vb12    84.6
## 32          asa24_vara    84.6
## 33           asa24_ret    84.6
## 34          asa24_bcar    84.6
## 35          asa24_acar    84.6
## 36          asa24_cryp    84.6
## 37          asa24_lyco    84.6
## 38            asa24_lz    84.6
## 39          asa24_atoc    84.6
## 40            asa24_vk    84.6
## 41         asa24_chole    84.6
## 42          asa24_sfat    84.6
## 43          asa24_s040    84.6
## 44          asa24_s060    84.6
## 45          asa24_s080    84.6
## 46          asa24_s100    84.6
## 47          asa24_s120    84.6
## 48          asa24_s140    84.6
## 49          asa24_s160    84.6
## 50          asa24_s180    84.6
## 51          asa24_mfat    84.6
## 52          asa24_m161    84.6
## 53          asa24_m181    84.6
## 54          asa24_m201    84.6
## 55          asa24_m221    84.6
## 56          asa24_pfat    84.6
## 57          asa24_p182    84.6
## 58          asa24_p183    84.6
## 59          asa24_p184    84.6
## 60          asa24_p204    84.6
## 61          asa24_p205    84.6
## 62          asa24_p225    84.6
## 63          asa24_p226    84.6
## 64          asa24_vitd    84.6
## 65         asa24_choln    84.6
## 66      asa24_vite_add    84.6
## 67       asa24_b12_add    84.6
## 68       asa24_f_total    84.6
## 69      asa24_f_citmlb    84.6
```

```
## 70             asa24_f_other     84.6
## 71             asa24_f_juice     84.6
## 72             asa24_v_total     84.6
## 73             asa24_v_drkgr     84.6
## 74       asa24_v_redor_total     84.6
## 75      asa24_v_redor_tomato     84.6
## 76       asa24_v_redor_other     84.6
## 77     asa24_v_starchy_total     84.6
## 78    asa24_v_starchy_potato     84.6
## 79     asa24_v_starchy_other     84.6
## 80             asa24_v_other     84.6
## 81           asa24_v_legumes     84.6
## 82             asa24_g_total     84.6
## 83             asa24_g_whole     84.6
## 84           asa24_g_refined     84.6
## 85            asa24_pf_total     84.6
## 86        asa24_pf_mps_total     84.6
## 87             asa24_pf_meat     84.6
## 88        asa24_pf_curedmeat     84.6
## 89            asa24_pf_organ     84.6
## 90            asa24_pf_poult     84.6
## 91         asa24_pf_seafd_hi     84.6
## 92        asa24_pf_seafd_low     84.6
## 93             asa24_pf_eggs     84.6
## 94              asa24_pf_soy     84.6
## 95           asa24_pf_nutsds     84.6
## 96          asa24_pf_legumes     84.6
## 97             asa24_d_total     84.6
## 98              asa24_d_milk     84.6
## 99            asa24_d_yogurt     84.6
## 100           asa24_d_cheese     84.6
## 101               asa24_oils     84.6
## 102         asa24_solid_fats     84.6
## 103         asa24_add_sugars     84.6
## 104           asa24_a_drinks     84.6
## 105          asa24_datacomp     84.6
## 106                 id_event      0
## 107                       id      0
## 108                    event      0
## 109                    study      0
```

interpretation :

- Notice the only features without missingness
- Notice the large amount of missingness is the

Participant info/Demographics

  - **`data_category`**

- Notice that the missingness appears to repeat – showing *runs* of the same `pct_miss` for both features

4.3 Ct and Graph Repeated Missingness

4.4 Diet Data

```
ct <-
  diet |>
  miss_var_summary() |>
  mutate(
    ct = ifelse(pct_miss == 0, 0, 1)
    ) |> count(ct)
ct
```

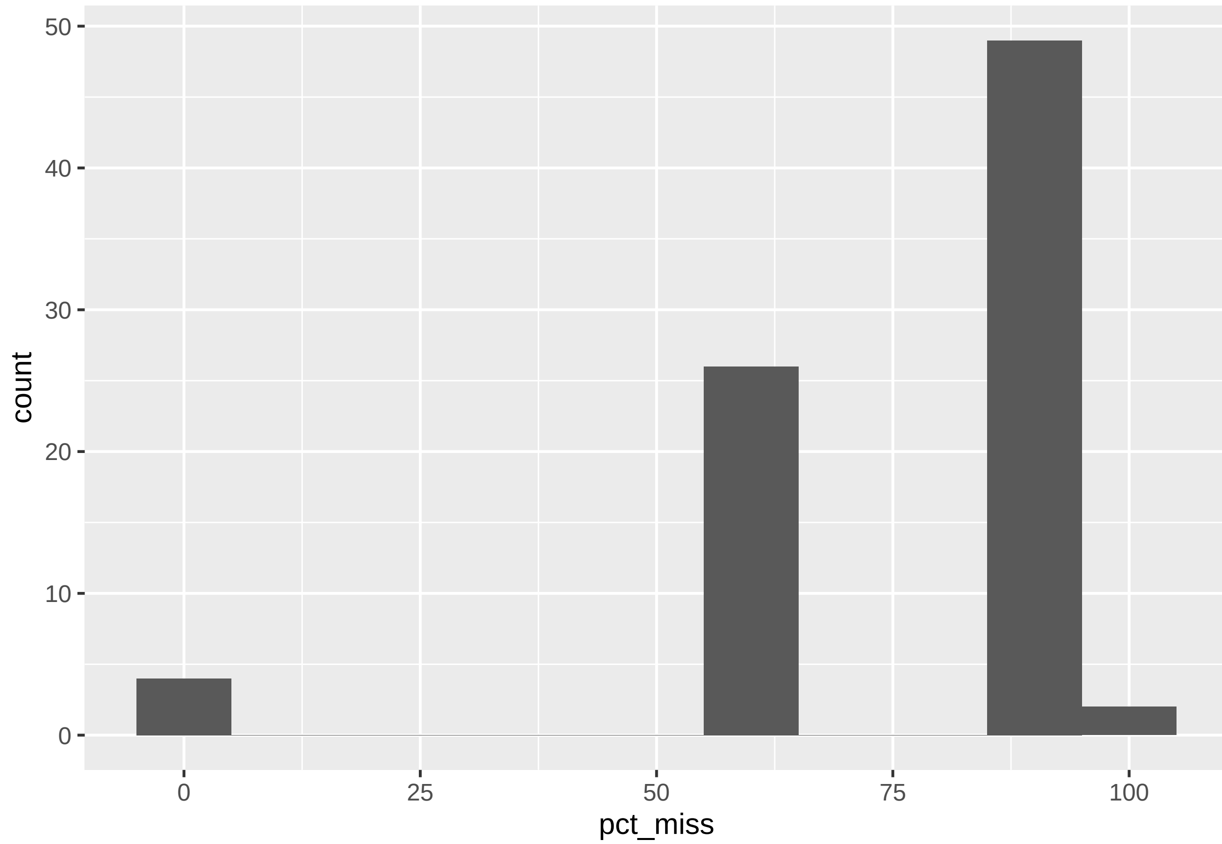
```
## # A tibble: 2 x 2
##      ct     n
##   <dbl> <int>
## 1     0     4
## 2     1   105
```

Interpretation :

- There are only 4 non-missing values every other value is just `84.6` percent missing – which is a **HUGE** amt.

4.5 Blood Data

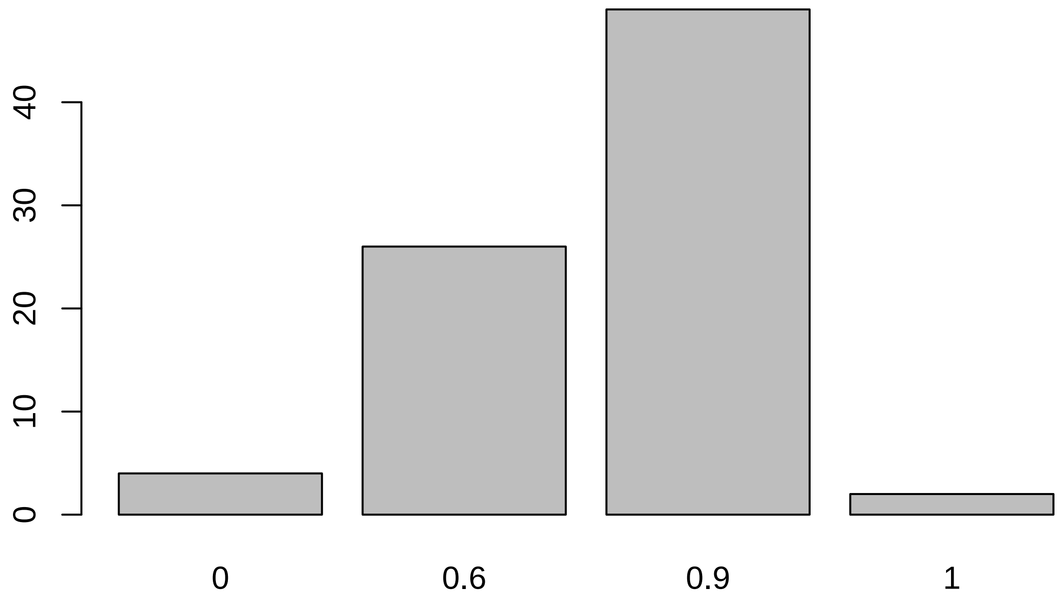
```
ct <-
  blood |>
  miss_var_summary()

library(ggplot2)
ct |>
  ggplot(aes(pct_miss)) +
  geom_histogram(binwidth = 10)
```



interpretation

- this one is slightly more varied
- 0, about 60, 90 100% percent missing – with 60, 90 happening quite frequently

```
ct <- ct |>
  mutate(
    char = format(pct_miss)
  )

ct <- as.vector(ct$pct_miss)/100
ct <- ct |> round(1)
ct <- ct |> as.character()
barplot(table(ct))
```



- We can see that the repeated values are near 0 (id features), 60%, 90% or 100%

5 Build Features

5.1 `patient` study & events ct

Purpose : identify how often a `patients` appear in our data.

For example :

- Are they in both studies?
- How many unique events?

5.1.1 Helper Func : Create `unique_id`, `Patient` Variable

```
library(dplyr)

diet <-
  diet |>
  mutate(
    patient = str_remove(id, "^[A-Z]+")
```

```
  ) |> select(patient, everything())

blood <-
  blood |>
  mutate(
    patient = str_remove(id, "^[A-Z]+")
  ) |> select(patient, everything())
```

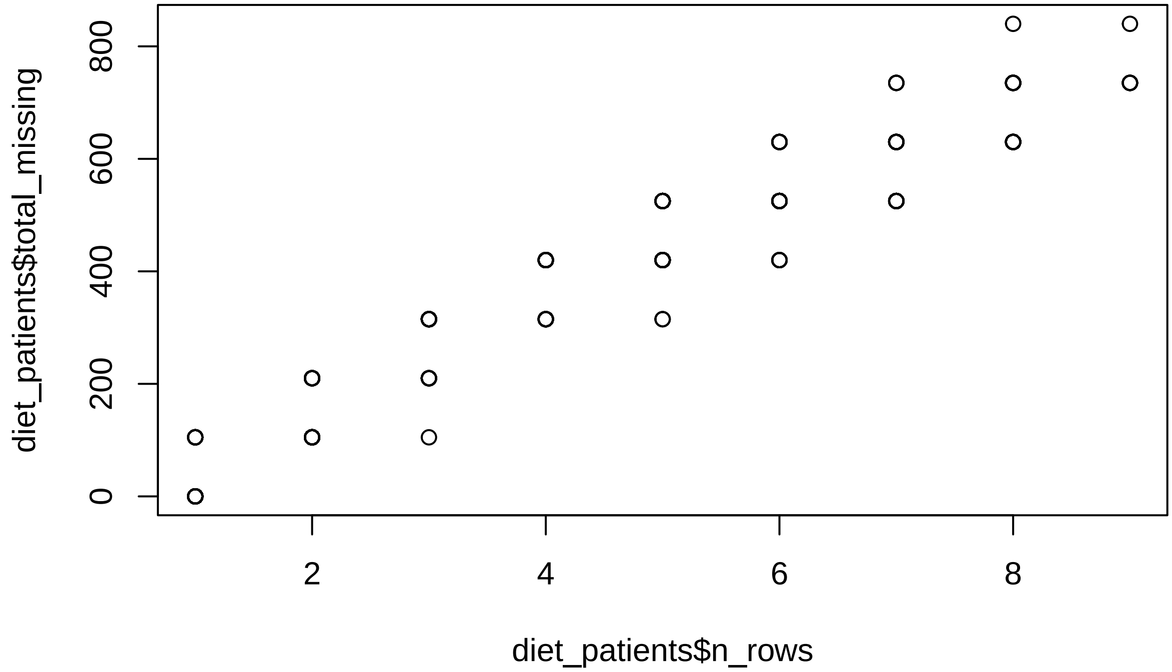
6 `patient` specific missing ct

Purpose : identify how often `patients` appear in our data to better understand missing values.

6.1 Diet

```
diet_patients <-
  diet |>
  group_by(patient) |>
  summarize(
    n_rows = n(),
    total_missing =
      sum(across(everything(), ~ sum(is.na(.x)))),
    .groups = "drop"
  ) |>
  mutate(
    pct =
      round(
        total_missing /
          (n_rows * (ncol(diet) - 1)),
        2
      )
  ) |>
  arrange(desc(pct))

diet_patients <-
  diet_patients |>  mutate(
    p_rank = paste0("patient", row_number())
  ) |> select(p_rank, everything(), patient)
```

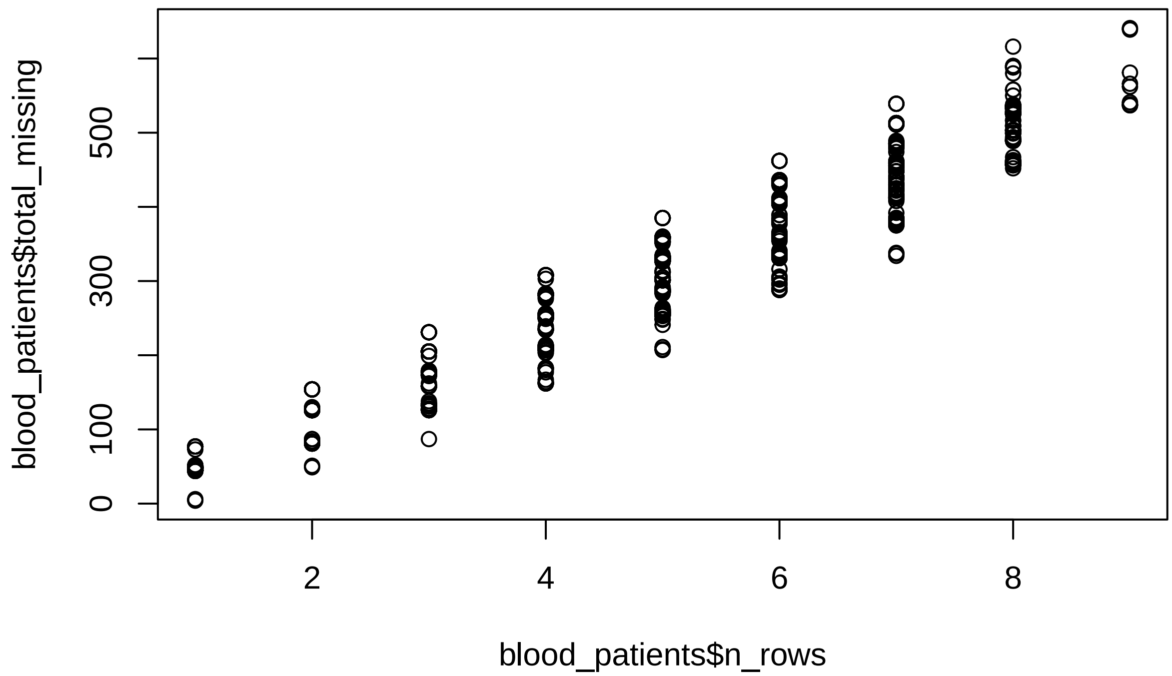
```
plot(diet_patients$n_rows, diet_patients$total_missing)
```



6.2 blood

```
blood_patients <-
  blood |>
  group_by(patient) |>
  summarize(
    n_rows = n(),
    total_missing =
      sum(across(everything(), ~ sum(is.na(.x)))),
    total_cells =
      n_rows * ncol(pick(everything())),
    pct =
      round(total_missing / total_cells, 2),
    .groups = "drop"
  ) |>
  arrange(desc(pct))

blood_patients <-
  blood_patients |>  mutate(
    p_rank = paste0("patient", row_number())
  ) |> select(p_rank, everything(), patient)
```

```r
plot(blood_patients$n_rows,blood_patients$total_missing)
```



7 Correlations of `Cognitive` Features – ALL

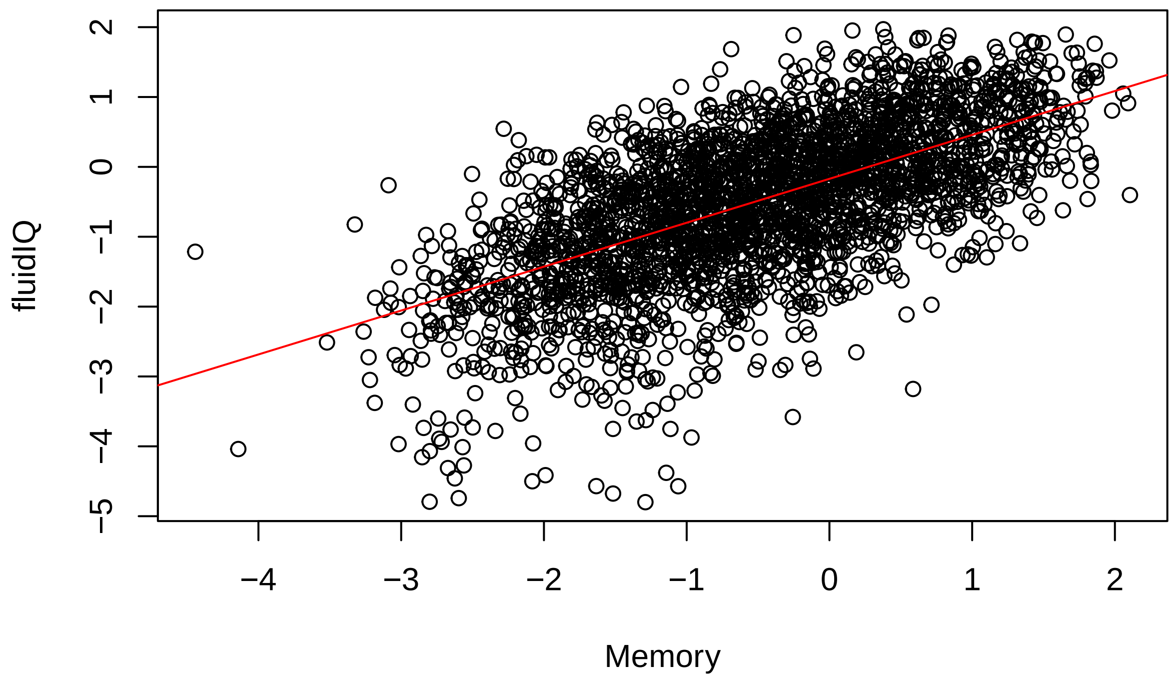
Note :

- we have repeated observations, so we are taking people who had more visits more into account
- Relevant factor analysis Features^
  - `Memory_Tr35_60y`,
    - Cognition Factor Analysis: Memory factor estimated in Union V1 35-60 yo subjects and applied to all subjects all visits
  - `FluidIQ_Tr35_60y`,
    - Cognition Factor Analysis: Fluid IQ factor estimated in Union V1 35-60 yo subjects and applied to all subjects all visits
  - `CrystIQ_Tr35_60y`
    - Cognition Factor Analysis: Crystallized IQ factor estimated in Union V1 35-60 yo subjects and applied to all subjects all visits

```r
Memory <- all$Memory_Tr35_60y |> as.numeric()
fluidIQ <- all$FluidIQ_Tr35_60y |> as.numeric()
CrystIQ <- all$CrystIQ_Tr35_60y |> as.numeric()
```

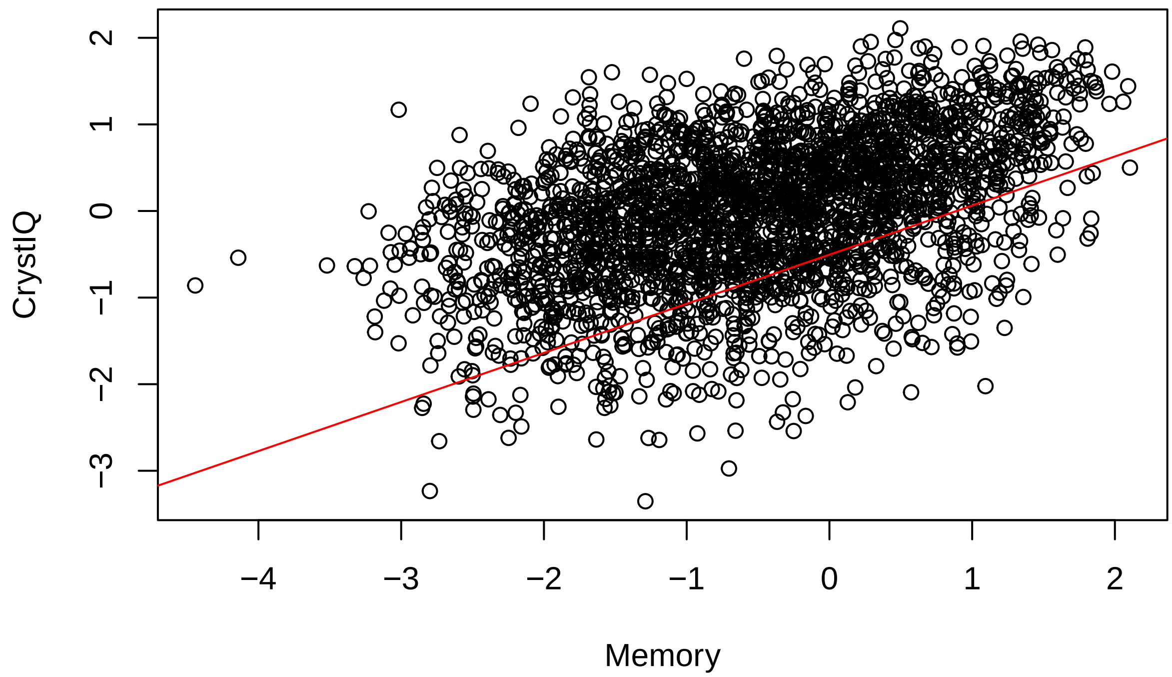
7.1 Memory, fluidIQ dot-plt

```r
# Memory ~ fluidIQ
Memory_fluidIQ_mdl <- lm(Memory ~ fluidIQ)

plot(Memory, fluidIQ)
abline(Memory_fluidIQ_mdl, col = "red")
```



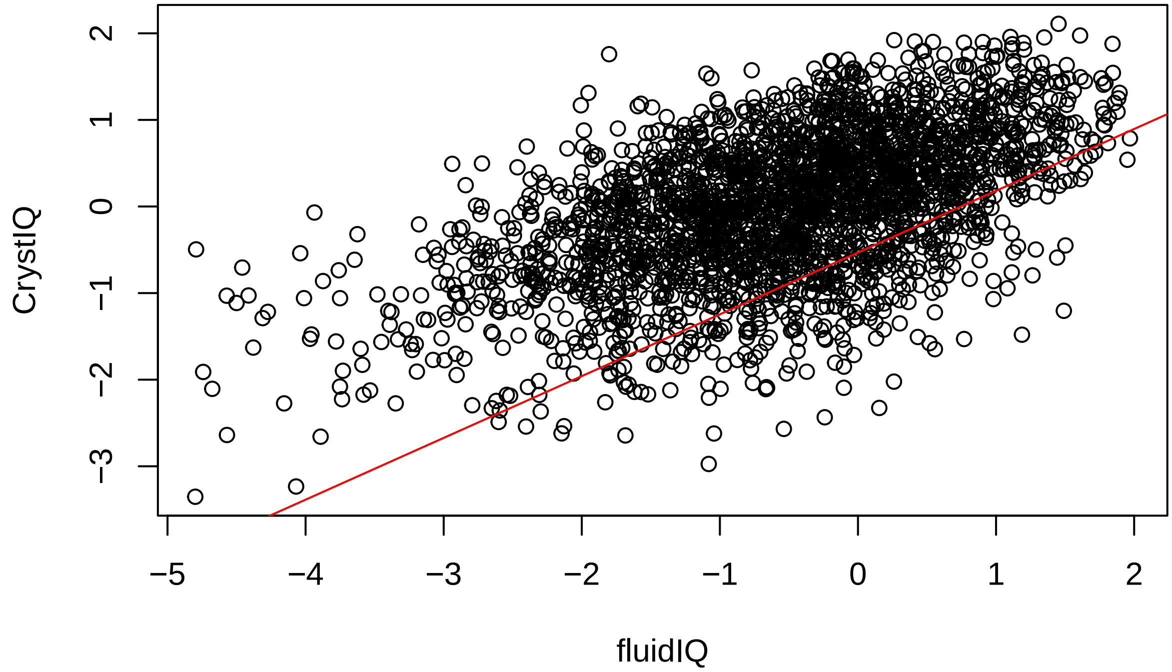
7.2 Memory, CrystIQ dot-plt

```r
# Memory ~ CrystIQ
Memory_CrystIQ_mdl <- lm(Memory ~ CrystIQ)

plot(Memory, CrystIQ)
abline(Memory_CrystIQ_mdl, col = "red")
```



7.3 fluidIQ, CrystIQ dot-plt

```
# fluidIQ ~ CrystIQ
fluidIQ_CrystIQ_mdl <- lm(fluidIQ ~ CrystIQ)

plot(fluidIQ, CrystIQ)
abline(fluidIQ_CrystIQ_mdl, col = "red")
```




```
diet_num <- diet |> select(where(is.numeric))
blood_num <- blood |> select(where(is.numeric))
```

```
diet_num <- cbind(Memory, fluidIQ, CrystIQ, diet_num)
blood_num <- cbind(Memory, fluidIQ, CrystIQ, blood_num)
```

8 Diet Correlations w/ `Memory`

```
mem_cor <-
  cor(diet_num$Memory, diet_num[4:length(diet_num)],
      use = "complete.obs")


mem_cor <-
  mem_cor |>
  t() |> as.data.frame()
```

```
mem_cor <-
  mem_cor |>
  rename(Memory = V1) |>
  round(3) |>
  select(Memory)
```

```
mem_cor
```

```
##                        Memory
## asa24_numfoods          0.156
## asa24_numcodes          0.136
## asa24_kcal              0.057
## asa24_prot              0.074
## asa24_tfat              0.064
## asa24_carb              0.040
## asa24_mois              0.156
## asa24_alc              -0.018
## asa24_caff              0.075
## asa24_theo              0.010
## asa24_sugr              0.010
## asa24_fibe              0.168
## asa24_calc              0.111
## asa24_iron              0.074
## asa24_magn              0.139
## asa24_phos              0.108
## asa24_pota              0.105
## asa24_sodi              0.043
## asa24_zinc              0.095
## asa24_copp              0.108
## asa24_sele              0.018
## asa24_vc                0.000
## asa24_vb1               0.084
## asa24_vb2               0.090
## asa24_niac              0.031
## asa24_vb6               0.040
## asa24_fola              0.119
## asa24_fa                0.038
## asa24_ff                0.148
## asa24_fdfe              0.097
## asa24_vb12              0.013
## asa24_vara              0.066
## asa24_ret               0.006
## asa24_bcar              0.100
## asa24_acar              0.057
## asa24_cryp             -0.012
## asa24_lyco             -0.030
## asa24_lz                0.121
## asa24_atoc              0.109
## asa24_vk                0.107
## asa24_chole            -0.010
## asa24_sfat              0.030
## asa24_s040              0.017
## asa24_s060              0.014
## asa24_s080              0.009
## asa24_s100              0.033
## asa24_s120             -0.008
## asa24_s140              0.016
## asa24_s160              0.036
## asa24_s180              0.021
```

```
## asa24_mfat              0.076
## asa24_m161             -0.001
## asa24_m181              0.080
## asa24_m201              0.022
## asa24_m221              0.018
## asa24_pfat              0.065
## asa24_p182              0.067
## asa24_p183              0.069
## asa24_p184             -0.006
## asa24_p204             -0.035
## asa24_p205             -0.038
## asa24_p225             -0.055
## asa24_p226             -0.049
## asa24_vitd             -0.005
## asa24_choln             0.026
## asa24_vite_add          0.044
## asa24_b12_add           0.042
## asa24_f_total           0.028
## asa24_f_citmlb          0.004
## asa24_f_other           0.102
## asa24_f_juice          -0.125
## asa24_v_total           0.096
## asa24_v_drkgr           0.117
## asa24_v_redor_total     0.057
## asa24_v_redor_tomato    0.018
## asa24_v_redor_other     0.077
## asa24_v_starchy_total  -0.028
## asa24_v_starchy_potato -0.037
## asa24_v_starchy_other   0.021
## asa24_v_other           0.085
## asa24_v_legumes         0.066
## asa24_g_total           0.022
## asa24_g_whole           0.018
## asa24_g_refined         0.016
## asa24_pf_total          0.023
## asa24_pf_mps_total     -0.065
## asa24_pf_meat          -0.019
## asa24_pf_curedmeat     -0.033
## asa24_pf_organ          0.025
## asa24_pf_poult         -0.003
## asa24_pf_seafd_hi      -0.033
## asa24_pf_seafd_low     -0.069
## asa24_pf_eggs           0.007
## asa24_pf_soy            0.077
## asa24_pf_nutsds         0.115
## asa24_pf_legumes        0.066
## asa24_d_total           0.093
## asa24_d_milk            0.051
## asa24_d_yogurt          0.097
## asa24_d_cheese          0.054
## asa24_oils              0.089
## asa24_solid_fats        0.002
## asa24_add_sugars       -0.019
## asa24_a_drinks         -0.017
```

```
## asa24_datacomp          0.037
```

9 Diet Correlations w/ `fluidIQ`

```
fluid_cor <-
  cor(diet_num$fluidIQ, diet_num[4:length(diet_num)],
      use = "complete.obs")


fluid_cor <-
  fluid_cor |>
  t() |> as.data.frame()

fluid_cor |>
  rename(fluidIQ = V1) |>
  round(3) |>
  select(fluidIQ)
```

```
##                         fluidIQ
## asa24_numfoods            0.158
## asa24_numcodes            0.138
## asa24_kcal                0.131
## asa24_prot                0.152
## asa24_tfat                0.125
## asa24_carb                0.086
## asa24_mois                0.161
## asa24_alc                 0.058
## asa24_caff                0.068
## asa24_theo                0.026
## asa24_sugr                0.023
## asa24_fibe                0.191
## asa24_calc                0.123
## asa24_iron                0.128
## asa24_magn                0.180
## asa24_phos                0.170
## asa24_pota                0.157
## asa24_sodi                0.117
## asa24_zinc                0.155
## asa24_copp                0.153
## asa24_sele                0.101
## asa24_vc                  0.012
## asa24_vb1                 0.115
## asa24_vb2                 0.144
## asa24_niac                0.120
## asa24_vb6                 0.097
## asa24_fola                0.171
## asa24_fa                  0.085
## asa24_ff                  0.176
## asa24_fdfe                0.149
## asa24_vb12                0.055
## asa24_vara                0.035
```

```
## asa24_ret                  0.020
## asa24_bcar                 0.026
## asa24_acar                 0.054
## asa24_cryp                -0.032
## asa24_lyco                 0.009
## asa24_lz                   0.056
## asa24_atoc                 0.140
## asa24_vk                   0.048
## asa24_chole                0.044
## asa24_sfat                 0.093
## asa24_s040                 0.053
## asa24_s060                 0.046
## asa24_s080                 0.054
## asa24_s100                 0.063
## asa24_s120                 0.041
## asa24_s140                 0.062
## asa24_s160                 0.099
## asa24_s180                 0.077
## asa24_mfat                 0.136
## asa24_m161                 0.055
## asa24_m181                 0.138
## asa24_m201                 0.085
## asa24_m221                 0.041
## asa24_pfat                 0.111
## asa24_p182                 0.116
## asa24_p183                 0.086
## asa24_p184                 0.013
## asa24_p204                 0.020
## asa24_p205                -0.019
## asa24_p225                -0.034
## asa24_p226                -0.019
## asa24_vitd                 0.013
## asa24_choln                0.094
## asa24_vite_add             0.036
## asa24_b12_add              0.062
## asa24_f_total              0.022
## asa24_f_citmlb            -0.012
## asa24_f_other              0.088
## asa24_f_juice             -0.093
## asa24_v_total              0.093
## asa24_v_drkgr              0.065
## asa24_v_redor_total        0.056
## asa24_v_redor_tomato       0.044
## asa24_v_redor_other        0.039
## asa24_v_starchy_total     -0.009
## asa24_v_starchy_potato    -0.012
## asa24_v_starchy_other      0.005
## asa24_v_other              0.095
## asa24_v_legumes            0.050
## asa24_g_total              0.078
## asa24_g_whole              0.038
## asa24_g_refined            0.066
## asa24_pf_total             0.112
## asa24_pf_mps_total         0.018
```

```
## asa24_pf_meat            0.004
## asa24_pf_curedmeat      -0.009
## asa24_pf_organ          -0.046
## asa24_pf_poult           0.055
## asa24_pf_seafd_hi       -0.007
## asa24_pf_seafd_low      -0.031
## asa24_pf_eggs            0.035
## asa24_pf_soy             0.054
## asa24_pf_nutsds          0.152
## asa24_pf_legumes         0.050
## asa24_d_total            0.109
## asa24_d_milk             0.063
## asa24_d_yogurt           0.100
## asa24_d_cheese           0.071
## asa24_oils               0.124
## asa24_solid_fats         0.050
## asa24_add_sugars        -0.005
## asa24_a_drinks           0.058
## asa24_datacomp           0.051
```

10 Diet Correlations w/ CrystIQ

```
cryst_cor <-
  cor(diet_num$CrystIQ, diet_num[4:length(diet_num)],
      use = "complete.obs")


cryst_cor <-
  cryst_cor |>
  t() |> as.data.frame()

cryst_cor <-
  cryst_cor |>
  rename(CrystIQ = V1) |>
  round(3) |> select(CrystIQ)


cryst_cor
```

```
##                        CrystIQ
## asa24_numfoods           0.188
## asa24_numcodes           0.160
## asa24_kcal               0.101
## asa24_prot               0.082
## asa24_tfat               0.095
## asa24_carb               0.083
## asa24_mois               0.108
## asa24_alc                0.062
## asa24_caff               0.102
## asa24_theo               0.056
## asa24_sugr               0.064
```

```
## asa24_fibe              0.214
## asa24_calc              0.123
## asa24_iron              0.140
## asa24_magn              0.175
## asa24_phos              0.147
## asa24_pota              0.159
## asa24_sodi              0.058
## asa24_zinc              0.122
## asa24_copp              0.167
## asa24_sele              0.023
## asa24_vc                0.024
## asa24_vb1               0.116
## asa24_vb2               0.149
## asa24_niac              0.040
## asa24_vb6               0.059
## asa24_fola              0.163
## asa24_fa                0.073
## asa24_ff                0.178
## asa24_fdfe              0.140
## asa24_vb12              0.067
## asa24_vara              0.097
## asa24_ret               0.039
## asa24_bcar              0.105
## asa24_acar              0.092
## asa24_cryp             -0.013
## asa24_lyco             -0.001
## asa24_lz                0.096
## asa24_atoc              0.112
## asa24_vk                0.068
## asa24_chole            -0.029
## asa24_sfat              0.083
## asa24_s040              0.055
## asa24_s060              0.051
## asa24_s080              0.068
## asa24_s100              0.061
## asa24_s120              0.059
## asa24_s140              0.067
## asa24_s160              0.077
## asa24_s180              0.075
## asa24_mfat              0.106
## asa24_m161              0.009
## asa24_m181              0.108
## asa24_m201              0.051
## asa24_m221              0.052
## asa24_pfat              0.069
## asa24_p182              0.075
## asa24_p183              0.056
## asa24_p184             -0.013
## asa24_p204             -0.050
## asa24_p205             -0.025
## asa24_p225             -0.047
## asa24_p226             -0.035
## asa24_vitd              0.022
## asa24_choln             0.040
```

```
## asa24_vite_add           0.037
## asa24_b12_add            0.076
## asa24_f_total            0.060
## asa24_f_citmlb           0.030
## asa24_f_other            0.106
## asa24_f_juice           -0.084
## asa24_v_total            0.080
## asa24_v_drkgr            0.085
## asa24_v_redor_total      0.098
## asa24_v_redor_tomato     0.057
## asa24_v_redor_other      0.095
## asa24_v_starchy_total   -0.021
## asa24_v_starchy_potato  -0.024
## asa24_v_starchy_other    0.004
## asa24_v_other            0.045
## asa24_v_legumes          0.066
## asa24_g_total            0.054
## asa24_g_whole            0.080
## asa24_g_refined          0.024
## asa24_pf_total           0.026
## asa24_pf_mps_total      -0.088
## asa24_pf_meat           -0.033
## asa24_pf_curedmeat      -0.017
## asa24_pf_organ          -0.042
## asa24_pf_poult          -0.038
## asa24_pf_seafd_hi       -0.014
## asa24_pf_seafd_low      -0.068
## asa24_pf_eggs           -0.008
## asa24_pf_soy             0.070
## asa24_pf_nutsds          0.159
## asa24_pf_legumes         0.066
## asa24_d_total            0.117
## asa24_d_milk             0.098
## asa24_d_yogurt           0.136
## asa24_d_cheese           0.047
## asa24_oils               0.072
## asa24_solid_fats         0.065
## asa24_add_sugars         0.017
## asa24_a_drinks           0.063
## asa24_datacomp           0.000
```

Overall interpretation

- Extremely weak correlations for all variables in Diet Data
    - Diet Data doesnt appear strongly related

Possible Causes

- underlying relationship isnt linear
- Missing values may be biasing variables

```r
diet_cor_lst <-
  list(
  cryst_cor=cryst_cor,
  fluid_cor=fluid_cor,
  mem_cor=mem_cor
)
```

11 Patient level analysis

```r
blood |>
  group_by(patient) |>
  select(patient, where(is.numeric)) |>
  summarize(
    across(everything(), ~ mean(.x, na.rm = TRUE))
  )
```

```
## # A tibble: 1,396 x 77
##    patient  hba1c  hscrp insulin vitamind albumin alkphos_total alt_sgpt
##    <chr>    <dbl>  <dbl>   <dbl>    <dbl>   <dbl>         <dbl>    <dbl>
##  1 6000030 NaN    NaN     NaN       NaN    NaN           NaN      NaN
##  2 6002236   5.03   4.63   18.0      84.1    4.13          61.3     27.3
##  3 6003238   6.1    1.13   27.4      41.6    4.15          50       24
##  4 6005242   8.55   3.12   10.0      26.4    4.15         112.      22.5
##  5 6007044   5.32   1.48   22.0      18.0    4.47          89.8     21.2
##  6 6010538   7.2    2.61    0.6      80.5    4.4          104       20
##  7 6012744   5.2  NaN      12        42.5    4.5           85       13
##  8 6016146   5.57   1.59    7.77     24.2    4.33          49       14.3
##  9 6018857   5.13   9.12   17.2      16.1    4.27          88       19.7
## 10 6030645   5.8    7.33   28.4      23.3    4.5           98       20
## # i 1,386 more rows
## # i 69 more variables: ast_sgot <dbl>, calcium <dbl>, chloride <dbl>,
## #   co2content <dbl>, creatinine <dbl>, glucose <dbl>, potassium <dbl>,
## #   sodium <dbl>, totalbilirubin <dbl>, totalprotein <dbl>, ureanitrogen <dbl>,
## #   friedewald_ldl <dbl>, hdl <dbl>, cholesterol <dbl>, triglyceride <dbl>,
## #   estradiol <dbl>, testosterone <dbl>, lh <dbl>, fsh <dbl>,
## #   aldosterone <dbl>, dheas <dbl>, cortisol <dbl>, CCL8 <dbl>, IL33 <dbl>, ...
```

12 Patient Correlations of `Cognitive` Features

- here we generate a report of correlations for each individual patient
- Suppose we just average out their values across all `Study` and `event`

```r
diet_patients
```

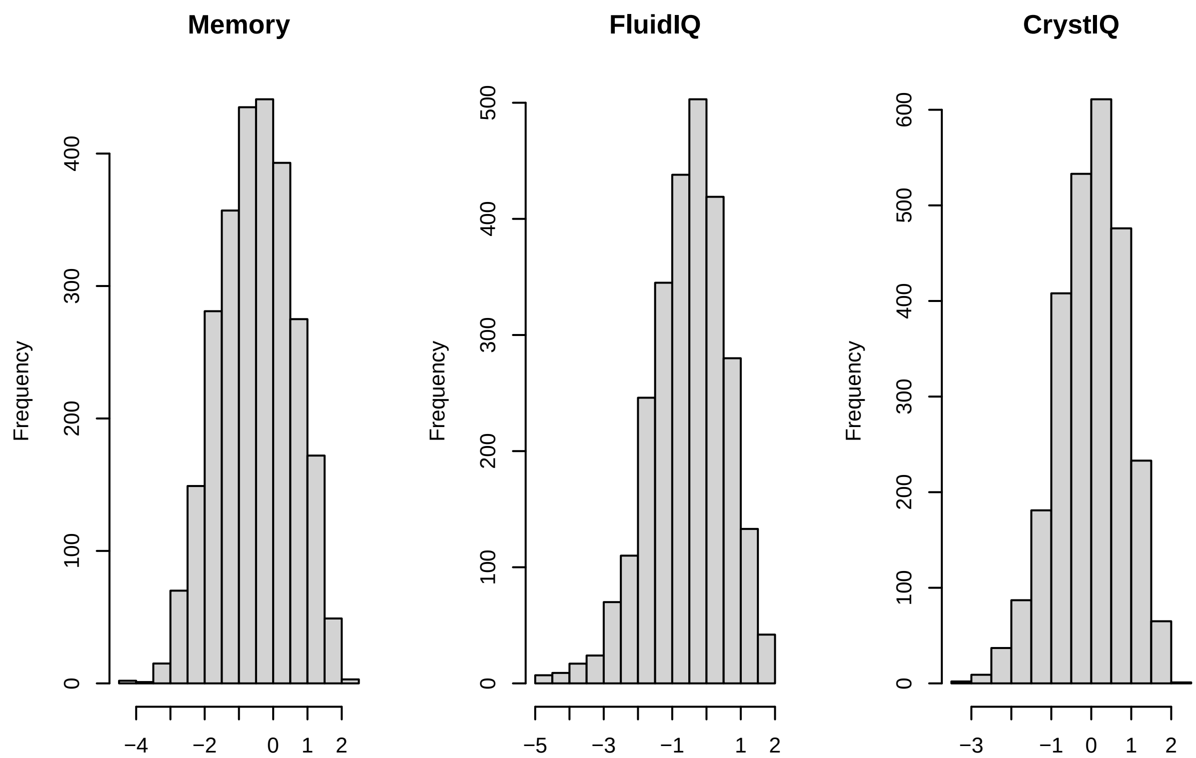
```
## # A tibble: 1,396 x 5
##    p_rank     patient n_rows total_missing   pct
##    <chr>      <chr>    <int>         <int> <dbl>
```

```
##  1 patient1  6000030      1            105  0.96
##  2 patient2  6012744      2            210  0.96
##  3 patient3  6030645      4            420  0.96
##  4 patient4  6053758      4            420  0.96
##  5 patient5  6058162      5            525  0.96
##  6 patient6  6061757      4            420  0.96
##  7 patient7  6075263      4            420  0.96
##  8 patient8  6086975      4            420  0.96
##  9 patient9  6089779      4            420  0.96
## 10 patient10 6098073      4            420  0.96
## # i 1,386 more rows
```

13 Multivariate Normal – Cognitive Features

```
cognitive_df <-
  all |>
  select(id_event, id, event, study,
         Memory_Tr35_60y,
         FluidIQ_Tr35_60y,
         CrystIQ_Tr35_60y) |>
  mutate(
    patient = str_remove(id, "^[A-Z]+")
  ) |> select(patient, everything()) |>
  mutate(
    across(
      -c(patient, id_event, id, event, study),
      as.numeric
    )
  )
```

```
par(mfrow = c(1, 3))

hist(
  cognitive_df$Memory_Tr35_60y,
  main = "Memory",
  xlab = ""
)

hist(
  cognitive_df$FluidIQ_Tr35_60y,
  main = "FluidIQ",
  xlab = ""
)

hist(
  cognitive_df$CrystIQ_Tr35_60y,
  main = "CrystIQ",
  xlab = ""
)
```



- notice each is normally distributed

```
cognitive_df |>
  select(where(is.numeric)) |>
  summarize(
    mean_Memory = mean(Memory_Tr35_60y, na.rm = TRUE),
    sd_Memory = sd(Memory_Tr35_60y, na.rm = TRUE),
    mean_FluidIQ_y = mean(FluidIQ_Tr35_60y, na.rm = TRUE),
    sd_FluidIQ = sd(FluidIQ_Tr35_60y, na.rm = TRUE),
    mean_CrystIQ = mean(CrystIQ_Tr35_60y, na.rm = TRUE),
    sd_CrystIQ = sd(CrystIQ_Tr35_60y, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 6
##   mean_Memory sd_Memory mean_FluidIQ_y sd_FluidIQ mean_CrystIQ sd_CrystIQ
##         <dbl>     <dbl>          <dbl>      <dbl>        <dbl>      <dbl>
## 1      -0.507      1.08         -0.535       1.10     -0.00429      0.845
```